Lecture 8

Induction

Objectives. *Lecture 7 found every irreducible representation of G inside $\mathbb{C}[G]$. Now a representation W of a subgroup H* induces *a representation of G, built from $\mathbb{C}[G]$ and W. Frobenius reciprocity relates maps out of an induced representation to maps between representations of the subgroup. Mackey's formula restricts an induced representation to a second subgroup; together they decide when $\operatorname{Ind}_H^G W$ is irreducible. Inducing the trivial representation gives the permutation modules M^μ of S_n, whose characters Lecture 13 needs.*

By the end of this lecture you will be able to:

- Construct $\operatorname{Ind}_H^G W$ as $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$, describe it on a transversal, and compute its character.
- Prove Frobenius reciprocity, and use it to decompose induced representations.
- Prove Mackey's formula, and decide by Mackey's criterion whether a representation induced from an irreducible one is irreducible.
- Compute the character of a permutation module M^μ by counting fixed cosets, and decompose $M^{(2,1)}$, $M^{(3,1)}$ and $M^{(2,2)}$.

8.1 The coset picture

Throughout this lecture, G is finite and $H \leq G$. Representations are complex and finite-dimensional.

Definition 8.1 (Restriction, transversals)**.** The *restriction* $\operatorname{Res}_H^G V$ of a representation (ρ, V) of G is the representation $\rho|_H$ of H on V, with character $\chi_V|_H$. A *transversal* of G/H is a subset $R \subseteq G$ that contains exactly one element of each left coset xH.

As G is the disjoint union of the cosets rH ($r \in R$), each $x \in G$ is rh for exactly one $r \in R$ and $h \in H$. For $x = gr$ this gives, for $g \in G$ and $r \in R$,

$$gr = r'h \qquad \text{with unique } r' \in R,\ h \in H.$$

So $r \mapsto r'$ is the action of g on G/H, read on representatives, and $h = r'^{-1}gr$ is left over.

Example 8.2 (S_2 in S_3)**.** Let $H = S_2 = \{e, (1\,2)\}$, the stabiliser of 3. By orbit–stabiliser (Theorem 1.14), $xH \mapsto x(3)$ is a bijection onto $\{1, 2, 3\}$ respecting the action. The three cosets were listed in § 1.2. The element $(2\,3)$ is in the same coset as $(1\,3\,2)$, so $R = \{e, (1\,3), (2\,3)\}$ is a transversal, with $r(3) = 3, 1, 2$. For $g = (1\,2\,3)$,

$$\begin{aligned} (1\,2\,3)\,e &= (1\,3)\,(1\,2), \\ (1\,2\,3)\,(1\,3) &= (2\,3)\,e, \\ (1\,2\,3)\,(2\,3) &= e\,(1\,2). \end{aligned}$$

So g moves the representatives $e \mapsto (1\,3) \mapsto (2\,3) \mapsto e$, as it moves $3 \mapsto 1 \mapsto 2 \mapsto 3$, leaving $(1\,2)$, e, $(1\,2)$ in H.

The permutation representation $\mathbb{C}^{G/H}$ (Example 5.2), with basis e_{xH} and $g \cdot e_{rH} = e_{r'H}$, records only r'. The subscript on e_{xH} marks a basis vector, not the identity element e. Induction also keeps h: it replaces each line $\mathbb{C}e_{rH}$ by a copy $r \otimes W$ of a representation W of H, and lets g act by

$$g \cdot (r \otimes w) = r' \otimes h \cdot w.$$

§ 8.2 builds it without choosing R.

The subgroup H need not be normal: S_2 is not normal in S_3, so S_3/S_2 is not a group (Example 1.13), but induction uses only the action on G/H.

8.2 $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ and its character

The subalgebra $\mathbb{C}[H] \subseteq \mathbb{C}[G]$ is spanned by H. By Proposition 7.5, the representation W of H is a left $\mathbb{C}[H]$-module. Modules are left modules (Definition 7.4), except that below $\mathbb{C}[G]$ is also a right $\mathbb{C}[H]$-module, with right action $a \cdot b = ab$ for $a \in \mathbb{C}[G]$ and $b \in \mathbb{C}[H]$.

Definition 8.3 (The balanced tensor product)**.** Let W be a representation of H. The *balanced tensor product* $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ is the quotient of $\mathbb{C}[G] \otimes W$ by the subspace spanned by

$$ah \otimes w - a \otimes h \cdot w \qquad (a \in \mathbb{C}[G],\ h \in H,\ w \in W).$$

The class of $a \otimes w$ is still written $a \otimes w$, so $ah \otimes w = a \otimes h \cdot w$.

Definition 8.4 (Balanced map)**.** Let X be a complex vector space. A bilinear map $\beta : \mathbb{C}[G] \times W \to X$ is *balanced* if $\beta(ah, w) = \beta(a, h \cdot w)$ for all $a \in \mathbb{C}[G]$, $h \in H$ and $w \in W$.

By linearity, the same relation holds when h is replaced by any element of $\mathbb{C}[H]$. The linear map $\mathbb{C}[G] \otimes W \to X$ induced by β vanishes on the subspace in Definition 8.3. Thus it gives a well-defined linear map

$$\overline{\beta} : \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W \to X, \qquad a \otimes w \longmapsto \beta(a, w).$$

Left multiplication by $g \in G$ maps the subspace to itself, as

$$g(ah) \otimes w - ga \otimes h \cdot w$$

is again a spanning vector. So $g \cdot (a \otimes w) = ga \otimes w$ is well defined, and makes $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ a $\mathbb{C}[G]$-module, a representation of G (Proposition 7.5).

Definition 8.5 (Induction)**.** The *induced representation* is $\operatorname{Ind}_H^G W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$.

Theorem 8.6 (Induced representations)**.** *Let W be a representation of H, and R a transversal of G/H.*

(i) Each element of $\operatorname{Ind}_H^G W$ is $\sum_{r \in R} r \otimes w_r$ for unique $w_r \in W$. So

$$\operatorname{Ind}_H^G W = \bigoplus_{r \in R} r \otimes W,$$

and $\dim \operatorname{Ind}_H^G W = [G : H] \dim W$.

(ii) If $gr = r'h$ with $r' \in R$ and $h \in H$, then $g \cdot (r \otimes w) = r' \otimes h \cdot w$.

(iii) The character of $\operatorname{Ind}_H^G W$ *is*

$$\chi(g) = \sum_{r\in R,\ r^{-1}gr\in H} \chi_W(r^{-1}gr)$$
$$= \frac{1}{|H|} \sum_{x\in G,\ x^{-1}gx\in H} \chi_W(x^{-1}gx).$$

Proof. (i) *Spanning.* $\mathbb{C}[G]\otimes W$ is spanned by the $x\otimes w$ with $x\in G$. Write $x = rh$; then $x\otimes w = r\otimes h\cdot w$.

Uniqueness. For $x = rh$, let $\beta(x,w)\in\bigoplus_{r\in R} W$ be $h\cdot w$ in the slot of r and 0 elsewhere, linear in x. It is balanced, as $xh' = r(hh')$ for $h'\in H$. So it defines a linear map sending $\sum_r r\otimes w_r$ to $(w_r)_{r\in R}$, which is 0 only if every $w_r = 0$.

(ii) As $g\cdot(r\otimes w) = gr\otimes w$,

$$g\cdot(r\otimes w) = r'h\otimes w = r'\otimes h\cdot w.$$

(iii) By (ii), g maps $r\otimes W$ onto $r'\otimes W$, so only the summands with $r' = r$, that is $r^{-1}gr\in H$, contribute to the trace. On them g acts as $r\otimes w\mapsto r\otimes\rho_W(r^{-1}gr)w$, of trace $\chi_W(r^{-1}gr)$. This is the first form.

For the second form, write $x = rh$:

$$x^{-1}gx = h^{-1}(r^{-1}gr)h$$

lies in H exactly when $r^{-1}gr$ does, with the same value of the class function χ_W. So each r occurs for $|H|$ values of x. ∎

Example 8.7 (From C_3 to S_3)**.** Let $H = C_3 = \langle(1\,2\,3)\rangle$ and $\zeta = \exp(2\pi\mathrm{i}/3)$, and let χ be the one-dimensional representation with $\chi((1\,2\,3)) = \zeta$ (Example 5.18). Take $R = \{e, (1\,2)\}$. No transposition is conjugate into C_3, so the induced character vanishes on transpositions. At $(1\,2\,3)$ both r contribute, as

$$(1\,2)^{-1}(1\,2\,3)(1\,2) = (1\,3\,2),$$

and the value there is

$$\chi\big((1\,2\,3)\big) + \chi\big((1\,3\,2)\big) = \zeta + \zeta^2 = -1.$$

Its degree is 2 by the dimension formula. The induced character is a class function, so its value on $(1\,3\,2)$ is also -1. These values identify it as χ_E (Example 6.4). So $\operatorname{Ind}_{C_3}^{S_3}\chi\cong E$, by Corollary 6.8(iii).

Example 8.8 (Guard: the representative matters)**.** The summand $r\otimes W$ depends only on the coset rH, since $rh\otimes W = r\otimes h\cdot W$. The vector $r\otimes w$ depends on r: changing the representative g to $g' = gh$ forces w to $w' = h^{-1}\cdot w$,

$$g\otimes w = gh\otimes h^{-1}\cdot w.$$

In Example 8.7, $(2\,3)$ lies in $(1\,2)H$, as

$$(1\,2)(1\,2\,3) = (2\,3),$$

and

$$(2\,3)\otimes w = (1\,2)\otimes(1\,2\,3)\cdot w = (1\,2)\otimes\zeta w,$$

not $(1\,2)\otimes w$.

8.3 Frobenius reciprocity

A G-map out of $\operatorname{Ind}_H^G W$ is determined on $e \otimes W$.

Theorem 8.9 (Frobenius reciprocity). *Let W be a representation of H, and V one of G.*

(i) The linear maps

$$\Phi\colon \operatorname{Hom}_G(\operatorname{Ind}_H^G W, V) \to \operatorname{Hom}_H(W, \operatorname{Res}_H^G V), \qquad \Phi(T)(w) = T(e \otimes w),$$

and

$$\Psi\colon \operatorname{Hom}_H(W, \operatorname{Res}_H^G V) \to \operatorname{Hom}_G(\operatorname{Ind}_H^G W, V), \qquad \Psi(f)(a \otimes w) = a \cdot f(w),$$

are well defined and inverse to each other.

(ii) The four numbers

$$\dim \operatorname{Hom}_G(\operatorname{Ind}_H^G W, V), \qquad \dim \operatorname{Hom}_G(V, \operatorname{Ind}_H^G W),$$
$$\dim \operatorname{Hom}_H(W, \operatorname{Res}_H^G V), \qquad \dim \operatorname{Hom}_H(\operatorname{Res}_H^G V, W)$$

are equal. In particular, for $U \in \operatorname{Irr}(G)$ and $W' \in \operatorname{Irr}(H)$, the multiplicity of U in $\operatorname{Ind}_H^G W'$ is that of W' in $\operatorname{Res}_H^G U$.

Proof. (i) *$\Psi(f)$ is well defined.* The map $(a, w) \mapsto a \cdot f(w)$ is bilinear, and balanced because f is an H-map:

$$(ah) \cdot f(w) = a \cdot \big(h \cdot f(w)\big) = a \cdot f(h \cdot w).$$

It is a G-map, as $\Psi(f)(ga \otimes w) = ga \cdot f(w)$.

$\Phi(T)$ is an H-map. The balanced relation gives $e \otimes h \cdot w = h \otimes w = h \cdot (e \otimes w)$. As T is a G-map,

$$\Phi(T)(h \cdot w) = T(e \otimes h \cdot w) = T\big(h \cdot (e \otimes w)\big) = h \cdot \Phi(T)(w).$$

Inverse maps. First,

$$\Phi(\Psi(f))(w) = e \cdot f(w) = f(w).$$

Second, T is a module map (Proposition 7.5(ii)), so

$$\Psi(\Phi(T))(a \otimes w) = a \cdot T(e \otimes w) = T(a \otimes w).$$

(ii) Reciprocity equates the first and third numbers. By Theorem 6.7,

$$\begin{aligned} \dim \operatorname{Hom}_G(A, B) &= \langle \chi_A, \chi_B \rangle_G \\ &= \overline{\langle \chi_B, \chi_A \rangle_G} = \overline{\dim \operatorname{Hom}_G(B, A)}, \end{aligned}$$

and a dimension is real; the same holds for H. Hence all four numbers agree. The multiplicities are the second and third numbers for $V = U$ and $W = W'$ (Corollary 5.16). ∎

Example 8.10 (Reciprocity for $C_3 \leq S_3$). We recover the induction in Example 8.7 using Frobenius reciprocity. On e, $(1\,2\,3)$ and $(1\,3\,2)$, the character of $\operatorname{Res}_{C_3}^{S_3} E$ takes the values $2, -1, -1$ (Example 6.4). These are the values of $\chi + \overline{\chi}$, since $\zeta + \zeta^2 = -1$. The two characters are distinct one-dimensional irreducible characters (Example 5.18). Their sum is the character of $\chi \oplus \overline{\chi}$, so by character orthogonality (Corollary 6.8), $\operatorname{Res}_{C_3}^{S_3} E \cong \chi \oplus \overline{\chi}$. Thus χ has multiplicity one in $\operatorname{Res}_{C_3}^{S_3} E$, and Frobenius reciprocity gives E multiplicity one in $\operatorname{Ind}_{C_3}^{S_3} \chi$. The induced representation has dimension 2, so it is E.

8.4 Mackey's formula and the irreducibility criterion

Let $K \leq G$ be a second subgroup. A conjugacy class is written $\mathcal{K}$ (Definition 7.10). What is $\operatorname{Res}^G_K \operatorname{Ind}^G_H W$?

Definition 8.11 (Double cosets, twisted representations). For $s \in G$, the *double coset* KsH is $\{ksh : k \in K,\ h \in H\}$. Let $K_s = K \cap sHs^{-1}$. For a representation W of H, the *twisted* representation ${}^sW$ of K_s is the space W with

$$({}^sW)(k) = \rho_W(s^{-1}ks) \qquad (k \in K_s).$$

The double cosets are the orbits of $K \times H$ on G under $(k,h) \cdot x = kxh^{-1}$, so they partition G, into the set $K\backslash G/H$.

The twisted space ${}^sW$ is a representation, as $k \mapsto s^{-1}ks$ is a homomorphism $K_s \to H$.

The double coset KsH is the union of the left cosets ksH, $k \in K$: the K-orbit of sH in G/H. Its stabiliser is

$$\{k \in K : ksH = sH\} = \{k \in K : s^{-1}ks \in H\} = K_s.$$

So, by orbit–stabiliser, KsH contains $[K : K_s]$ cosets of H.

Theorem 8.12 (Mackey's formula). *Let S be a set of representatives of $K\backslash G/H$. Then*

$$\operatorname{Res}^G_K \operatorname{Ind}^G_H W \cong \bigoplus_{s \in S} \operatorname{Ind}^K_{K_s} {}^sW.$$

Proof. Let $V = \operatorname{Ind}^G_H W$, and for $s \in S$ let $V(s)$ be the span of the $ks \otimes W$, $k \in K$.

A splitting. By Theorem 8.6(i), V is the direct sum of the spaces $x \otimes W$, one for each coset xH (Example 8.8), and $V(s)$ is the sum of those for the $[K : K_s]$ cosets ksH in KsH. Each coset lies in one double coset, so

$$V = \bigoplus_{s \in S} V(s),$$

and each $V(s)$, of dimension $[K : K_s] \dim W$, is K-stable, as $k' \cdot (ks \otimes w) = k'ks \otimes w$.

An intertwiner. For $k \in K_s$, $s^{-1}ks \in H$, so

$$k \cdot (s \otimes w) = s(s^{-1}ks) \otimes w = s \otimes \rho_W(s^{-1}ks)w.$$

So $w \mapsto s \otimes w$ is a K_s-map ${}^sW \to \operatorname{Res}^K_{K_s} V(s)$. By Frobenius reciprocity for $K_s \leq K$, the map

$$\operatorname{Ind}^K_{K_s} {}^sW \to V(s), \qquad a \otimes w \mapsto a \cdot (s \otimes w),$$

is a K-map.

An isomorphism. Its image contains every $ks \otimes W$, so it is onto, and by Theorem 8.6(i) both sides have dimension $[K : K_s] \dim W$. ∎

Example 8.13 (A non-normal subgroup). Let $H = K = S_2 = \{e, (1\,2)\} \leq S_3$. The double cosets are H and $H(1\,3)H$: they have two and four elements and together cover S_3. For the representative $(1\,3)$, the intersection is

$$H \cap (1\,3)H(1\,3)^{-1} = \{e\}.$$

Mackey's formula applied to the trivial representation gives

$$\operatorname{Res}^{S_3}_H \operatorname{Ind}^{S_3}_H \mathbf{1} \cong \mathbf{1} \oplus \operatorname{Ind}^H_{\{e\}} \mathbf{1} \cong \mathbf{1}^{\oplus 2} \oplus \operatorname{sgn}.$$

The second induced representation is $\mathbb{C}[H] \otimes_{\mathbb{C}} \mathbb{C}$, which is the regular representation $\mathbb{C}[H]$. Its two one-dimensional summands were found in § 7.2.

Example 8.14 (Guard: Mackey's twist)**.** Let $H = K = C_3 \leq S_3$ and $W = \chi$, as in Example 8.7. C_3 is normal, so its double cosets are its two cosets, with representatives e and $s = (1\,2)$, and $K_s = C_3$. Conjugation by s inverts C_3, for instance

$$(1\,2)^{-1}(1\,2\,3)(1\,2) = (1\,3\,2),$$

so

$${}^s\chi(k) = \chi(k^{-1}) = \overline{\chi(k)}.$$

Mackey's formula gives

$$\operatorname{Res}^{S_3}_{C_3} \operatorname{Ind}^{S_3}_{C_3} \chi \cong \chi \oplus \overline{\chi},$$

not 2χ. Since $\operatorname{Ind}^{S_3}_{C_3} \chi \cong E$, this agrees with the decomposition of $\operatorname{Res}^{S_3}_{C_3} E$ in Example 8.10.

Corollary 8.15 (Mackey's irreducibility criterion)**.** *Let W be an irreducible representation of H, and for $s \in G$ let $H_s = H \cap sHs^{-1}$. Then $\operatorname{Ind}^G_H W$ is irreducible if and only if, for every $s \in G \setminus H$, the representations ${}^sW$ and $\operatorname{Res}^H_{H_s} W$ of H_s have no irreducible constituent in common.*

Proof. A sum over double cosets. Let S represent $H\backslash G/H$, with e representing H. By Theorem 8.9(i), Theorem 8.12 with $K = H$, and Theorem 8.9(ii) for $H_s \leq H$,

$$\begin{aligned}
\dim \operatorname{End}_G(\operatorname{Ind}^G_H W) &= \dim \operatorname{Hom}_H(W, \operatorname{Res}^G_H \operatorname{Ind}^G_H W) \\
&= \sum_{s \in S} \dim \operatorname{Hom}_H\left(W, \operatorname{Ind}^H_{H_s} {}^sW\right) \\
&= \sum_{s \in S} \dim \operatorname{Hom}_{H_s}\left(\operatorname{Res}^H_{H_s} W, {}^sW\right).
\end{aligned}$$

The term of e. $H_e = H$ and ${}^eW = W$, so it is $\dim \operatorname{End}_H(W) = 1$, by Schur's lemma (Theorem 5.13(ii)).

The other terms. By Theorem 6.7 and Corollary 6.8(ii), with real multiplicities,

$$\dim \operatorname{Hom}_{H_s}(A, B) = \sum_{U \in \operatorname{Irr}(H_s)} m_U(A)\, m_U(B),$$

which is 0 exactly when A and B have no common constituent.

The count. By Theorem 7.17, a representation V of G has

$$\dim \operatorname{End}_G(V) = \sum_U m_U(V)^2,$$

which is 1 exactly when V is irreducible. Here $W \neq 0$, so the dimension formula gives $\operatorname{Ind}^G_H W \neq 0$. Thus $\operatorname{Ind}^G_H W$ is irreducible if and only if every term with $s \in S \setminus \{e\}$ vanishes. Each $s \notin H$ lies in a double coset other than H, and S may be chosen to contain it; so this holds for one S exactly when it holds for every $s \notin H$. ∎

Example 8.16 (Three induced planes)**.** Each H below is normal of index 2, so $H_s = H$.

- $C_3 \leq S_3$. Each $s \notin C_3$ is a transposition, with ${}^s\chi = \overline{\chi} \neq \chi$ (Example 8.14): E is irreducible.
- $C_4 \leq D_4$. Here r is the rotation, not a transversal element. Let $C_4 = \langle r \rangle$ and $\chi(r) = \mathrm{i}$. Every $s' \notin C_4$ is some $r^k s$ (Example 1.18), and

$$s'^{-1} r s' = s^{-1} r s = r^{-1},$$

so ${}^{s'}\chi(r) = -\mathrm{i}$, and $\mathrm{Ind}_{C_4}^{D_4}\chi$ is irreducible. On the transversal $\{e, s\}$, as $rs = sr^{-1}$, Theorem 8.6(ii) gives

$$r\cdot(e\otimes w) = \mathrm{i}\,(e\otimes w), \qquad r\cdot(s\otimes w) = -\mathrm{i}\,(s\otimes w),$$
$$s\cdot(e\otimes w) = s\otimes w, \qquad s\cdot(s\otimes w) = e\otimes w.$$

These are the columns of Lecture 5's matrices of r and s (§ 5.5): its plane is $\mathrm{Ind}_{C_4}^{D_4}\chi$.

- $\langle i\rangle \le Q_8$. The quaternion group and its subgroup are

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \qquad \langle i\rangle = \{\pm 1, \pm i\}$$

with

$$i^2 = -1, \qquad j^2 = -1, \qquad k^2 = -1, \qquad ijk = -1.$$

The identity is $e = 1$, distinct from the group element -1. The italic i, j, k are quaternions, and the upright i is the complex unit. The group element -1 is central and has order 2. Each of i, j and k has inverse its negative. Since $ijk = -1$ and $k^{-1} = -k$, we have $ij = k$. Inverting gives $ji = -k$, so i and j anticommute. Thus $j^{-1}ij = -i$. Since $k = ij$, conjugation by k gives the same result; centrality of -1 shows that signs do not change these conjugations.

Let $\chi(i) = \mathrm{i}$. Every $s \notin \langle i\rangle$ is $\pm j$ or $\pm k$, and $s^{-1}is = -i$, so ${}^{s}\chi(i) = -\mathrm{i}$, and $\mathrm{Ind}_{\langle i\rangle}^{Q_8}\chi$ is irreducible. On the transversal $\{e, j\}$, i acts as r did. Since $-1 = i^2$ lies in $\langle i\rangle$, $\chi(-1) = \chi(i)^2 = -1$. In the following calculation the first -1 is a group element. The balanced relation moves it to the second factor, where its action is the scalar $\chi(-1)$:

$$j\cdot(j\otimes w) = j^2\otimes w = (-1)\otimes w = e\otimes\chi(-1)w = -(e\otimes w).$$

Thus j acts with square $-\mathbb{1}$, while s acts with square $\mathbb{1}$ in the preceding D_4 representation.

In the D_4 case the choice of character matters: for $\chi(r) = -1$, ${}^{s}\chi = \chi$, and $\mathrm{Ind}_{C_4}^{D_4}\chi$ is reducible.

8.5 Permutation modules

Definition 8.17 (Young subgroup and permutation module)**.** For $\mu = (\mu_1, \dots, \mu_\ell) \vdash n$, the *Young subgroup* $S_\mu \le S_n$ consists of the permutations that preserve each block as a set:

$$B_i = \left\{1 + \sum_{j=1}^{i-1}\mu_j, \dots, \sum_{j=1}^{i}\mu_j\right\} \qquad (1 \le i \le \ell).$$

An empty sum is 0, so the first block starts at 1. The last block B_ℓ ends at n. The *permutation module* is $M^\mu = \mathrm{Ind}_{S_\mu}^{S_n}\mathbf{1}$.

The permutations on the blocks are independent, so $S_\mu \cong S_{\mu_1}\times\cdots\times S_{\mu_\ell}$.

Proposition 8.18 (Permutation modules)**.**

(i) $\mathrm{Ind}_H^G\mathbf{1} \cong \mathbb{C}^{G/H}$*, the permutation representation on the cosets, by* $r\otimes 1 \mapsto e_{rH}$*. Its character at* g *is the number of cosets* xH *with* $gxH = xH$*.*

(ii) Let X_μ *be the set of ordered set partitions* $(A_1, \dots, A_\ell)$ *of* $\{1, \dots, n\}$*: the blocks are pairwise disjoint, have union* $\{1, \dots, n\}$ *and satisfy* $|A_i| = \mu_i$*. The order of the blocks matters even when two have the same size. The group* S_n *acts on* X_μ *by* $\sigma\cdot(A_i)_i = (\sigma(A_i))_i$*. Then* $\mathrm{Ind}_{S_\mu}^{S_n}\mathbf{1} \cong \mathbb{C}^{X_\mu}$*, and its character at* σ *is the number of* $(A_i) \in X_\mu$ *with every* A_i *a union of cycles of* σ*.*

Proof. (i) By Theorem 8.6(ii) with $W = \mathbf{1}$, $g \cdot (r \otimes 1) = r' \otimes 1$, where $grH = r'H$. So $r \otimes 1 \mapsto e_{rH}$ is an isomorphism, and the character counts fixed points (Example 5.9).

(ii) *Transitive.* Two partitions in X_μ differ by a permutation that maps each A_i onto A'_i.

The stabiliser of $(B_1, \dots, B_\ell)$ is S_μ. So, by Theorem 1.14, $\sigma S_\mu \mapsto (\sigma(B_i))_i$ is a bijection $S_n/S_\mu \to X_\mu$ that carries the action across, and (i) gives $M^\mu \cong \mathbb{C}^{X_\mu}$.

The fixed points. $\sigma(A) = A$ exactly when A is a union of cycles of σ. ∎

For example, $X_{(2,2)}$ has six elements: choosing the first 2-element block determines the second. Forgetting the order identifies each pair with its reversal and gives the three pairings of § 2.6. Lecture 13 uses the fixed-coset character formula for all M^μ at once.

8.6 Worked example

We decompose the permutation modules $M^{(2,1)}$, $M^{(3,1)}$ and $M^{(2,2)}$.

$M^{(2,1)}$. $S_{(2,1)} = \{e, (1\,2)\}$ is the S_2 of Example 8.2, and $(\{a,b\},\{c\}) \mapsto c$ identifies $X_{(2,1)}$ with $\{1,2,3\}$. So $M^{(2,1)} \cong \mathbb{C}^3$, the representation of Lecture 5. By Theorem 8.9(ii), $U \in \mathrm{Irr}(S_3)$ occurs in it as often as $\mathbf{1}$ occurs in $\mathrm{Res}^{S_3}_{S_2} U$. By Example 7.21, $\mathbf{1}$, sgn and E restrict to S_2 as

$$\mathbf{1}, \qquad \mathrm{sgn}, \qquad \mathbf{1} \oplus \mathrm{sgn}\,.$$

So

$$M^{(2,1)} \cong \mathbf{1} \oplus E,$$

as Lecture 5 found by hand.

$M^{(3,1)}$. $S_{(3,1)}$ is the stabiliser of 4, a copy of S_3, and $M^{(3,1)} \cong \mathbb{C}^4$. Again U occurs as often as $\mathbf{1}$ in $\mathrm{Res}^{S_4}_{S_3} U$, which is $\langle 1_{S_3}, \chi_U \rangle_{S_3}$. Restrict the table of § 6.6 to S_3: its classes are those of e, $(1\,2)$ and $(1\,2\,3)$, of sizes 1, 3, 2, with the columns (1^4), $(2,1^2)$ and $(3,1)$. For the standard W, its sign twist and the lift $\rho_E \circ \pi$,

$$\begin{aligned} m_W &= \frac{1}{6}(3 + 3\cdot 1 + 2\cdot 0) = 1, \\ m_{\mathrm{sgn}\otimes W} &= \frac{1}{6}(3 - 3\cdot 1 + 2\cdot 0) = 0, \\ m_{\rho_E\circ\pi} &= \frac{1}{6}(2 + 3\cdot 0 - 2\cdot 1) = 0, \end{aligned}$$

and $\mathbf{1}$, sgn restrict to $\mathbf{1}$, sgn, with $m_{\mathbf{1}} = 1$, $m_{\mathrm{sgn}} = 0$. So

$$M^{(3,1)} \cong \mathbf{1} \oplus W,$$

recovering Lecture 6's splitting. Lecture 12 labels W by $(3,1)$.

Self-check. Decompose $M^{(2,2)}$. *Answer:* by Proposition 8.18(ii), $\chi(\sigma)$ counts the 2-sets A_1 that are unions of cycles of σ, A_2 being the complement. The identity fixes all six 2-sets. For a transposition, choose its cycle or the two fixed points; for two transpositions, choose either cycle. A 3- or 4-cycle leaves no such set. Thus

$$\begin{aligned} \langle 1_{S_4}, \chi \rangle_{S_4} &= \frac{1}{24}(6 + 6\cdot 2 + 3\cdot 2) = 1, \\ \langle \chi_W, \chi \rangle_{S_4} &= \frac{1}{24}(18 + 6\cdot 2 - 3\cdot 2) = 1, \\ \langle \chi_E \circ \pi, \chi \rangle_{S_4} &= \frac{1}{24}(12 + 0 + 3\cdot 4) = 1, \end{aligned}$$

while sgn and sgn $\otimes W$ give 0. So

$$M^{(2,2)} \cong \mathbf{1} \oplus W \oplus (\rho_E \circ \pi).$$

Summary.

- *The main results:* $\operatorname{Ind}_H^G W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ and its character (Theorem 8.6); Frobenius reciprocity (Theorem 8.9); Mackey's formula (Theorem 8.12) and irreducibility criterion for irreducible W (Corollary 8.15).
- *Handed on:* the permutation modules M^μ and their fixed-coset characters (Proposition 8.18) to Lecture 13; induction, reciprocity and Mackey to Sheets 8–9.
- *Next:* Lecture 9 asks what the values of characters know about the group.

References

Serre, *Linear Representations of Finite Groups*: induced representations, §3.3, pp. 28–31, with their character, Thm 12, p. 30; $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$, Prop. 18, p. 54; the induced character, Prop. 20, and Frobenius reciprocity, Thm 13, p. 56; restriction to subgroups, Prop. 22, p. 58; Mackey's criterion, Prop. 23, p. 59.[1] Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: induced representations, Def. 1.6.1, p. 51; permutation representations, Prop. 1.6.5, p. 52; the character, Thm 1.6.7, p. 53; reciprocity, Thm 1.6.11, p. 55; Mackey's lemma, Thm 1.6.14, p. 57.[2]

[1] Serre's $\rho^s(x) = \rho(s^{-1}xs)$ on $H_s = sHs^{-1} \cap K$ is our twist.
[2] Their H and K are our K and H.